

- $\mathbb{R}^n$ is a division algebra only for $n = 1, 2, 4, 8$.
- S^n is an H-space only for $n = 0, 1, 3, 7$.
- S^n has n linearly independent tangent vector fields only for $n = 0, 1, 3, 7$.
- The only fiber bundles $S^p \rightarrow S^q \rightarrow S^r$ occur when $(p, q, r) = (0, 1, 1), (1, 3, 2), (3, 7, 4)$, and $(7, 15, 8)$.

The first and third assertions were in fact proved shortly before Adams' theorem in [Kervaire 1958] and [Milnor 1958] as applications of a theorem of Bott that $\pi_{2n}U(n) \approx \mathbb{Z}_n$. A full discussion of all this, and a proof of Adams' theorem, is given in [VBKT].

Though maps of Hopf invariant 1 are rare, there are maps $S^{2n-1} \rightarrow S^n$ of Hopf invariant 2 for all even n . Namely, consider the space $J_2(S^n)$ constructed in §3.2. This has a CW structure with three cells, of dimensions 0, n , and $2n$, so $J_2(S^n)$ has the form C_f for some $f: S^{2n-1} \rightarrow S^n$. We showed that if n is even, the square of a generator of $H^n(J_2(S^n); \mathbb{Z})$ is twice a generator of $H^{2n}(J_2(S^n); \mathbb{Z})$, so $H(f) = \pm 2$.

From this example we can get maps of any even Hopf invariant when n is even via the following fact.

Proposition 4B.1. *The Hopf invariant $H: \pi_{2n-1}(S^n) \rightarrow \mathbb{Z}$ is a homomorphism.*

Proof: For $f, g: S^{2n-1} \rightarrow S^n$, let us compare C_{f+g} with the space $C_{f \vee g}$ obtained from S^n by attaching two $2n$ -cells via f and g . There is a natural quotient map $q: C_{f+g} \rightarrow C_{f \vee g}$ collapsing the equatorial disk of the $2n$ -cell of C_{f+g} to a point. The induced cellular chain map q_* sends e_{f+g}^{2n} to $e_f^{2n} + e_g^{2n}$. In cohomology this implies that $q^*(\beta_f) = q^*(\beta_g) = \beta_{f+g}$ where β_f, β_g , and β_{f+g} are the cohomology classes dual to the $2n$ -cells. Letting α_{f+g} and $\alpha_{f \vee g}$ be the cohomology classes corresponding to the n -cells, we have $q^*(\alpha_{f \vee g}) = \alpha_{f+g}$ since q is a homeomorphism on the n -cells. By restricting to the subspaces C_f and C_g of $C_{f \vee g}$ we see that $\alpha_{f \vee g}^2 = H(f)\beta_f + H(g)\beta_g$. Thus $\alpha_{f+g}^2 = q^*(\alpha_{f \vee g}^2) = H(f)q^*(\beta_f) + H(g)q^*(\beta_g) = (H(f) + H(g))\beta_{f+g}$. $\square$

Corollary 4B.2. *$\pi_{2n-1}(S^n)$ contains a $\mathbb{Z}$ direct summand when n is even.*

Proof: Either H or $H/2$ is a surjective homomorphism $\pi_{2n-1}(S^n) \rightarrow \mathbb{Z}$. $\square$

Exercises

1. Show that the Hopf invariant of a composition $S^{2n-1} \xrightarrow{f} S^{2n-1} \xrightarrow{g} S^n$ is given by $H(gf) = (\deg f)H(g)$, and for a composition $S^{2n-1} \xrightarrow{f} S^n \xrightarrow{g} S^n$ the Hopf invariant satisfies $H(gf) = (\deg g)^2 H(f)$.
2. Show that if $S^k \rightarrow S^m \xrightarrow{p} S^n$ is a fiber bundle, then $m = 2n - 1$, $k = n - 1$, and, when $n > 1$, $H(p) = \pm 1$. [Show that C_p is a manifold and apply Poincaré duality.]

4.C Minimal Cell Structures

We can apply the homology version of Whitehead's theorem, Corollary 4.33, to show that a simply-connected CW complex with finitely generated homology groups is always homotopy equivalent to a CW complex having the minimum number of cells consistent with its homology, namely, one n -cell for each $\mathbb{Z}$ summand of H_n and a pair of cells of dimension n and $n + 1$ for each $\mathbb{Z}_k$ summand of H_n .

Proposition 4C.1. *Given a simply-connected CW complex X and a decomposition of each of its homology groups $H_n(X)$ as a direct sum of cyclic groups with specified generators, then there is a CW complex Z and a cellular homotopy equivalence $f: Z \rightarrow X$ such that each cell of Z is either:*

- (a) a 'generator' n -cell e_α^n , which is a cycle in cellular homology mapped by f to a cellular cycle representing the specified generator α of one of the cyclic summands of $H_n(X)$; or
- (b) a 'relator' $(n + 1)$ -cell e_α^{n+1} , with cellular boundary equal to a multiple of the generator n -cell e_α^n , in the case that α has finite order.

In the nonsimply-connected case this result can easily be false, counterexamples being provided by acyclic spaces and the space $X = (S^1 \vee S^n) \cup e^{n+1}$ constructed in Example 4.35, which has the same homology as S^1 but which must have cells of dimension greater than 1 in order to have π_n nontrivial.

Proof: We build Z inductively over skeleta, starting with Z^1 a point since X is simply-connected. For the inductive step, suppose we have constructed $f: Z^n \rightarrow X$ inducing an isomorphism on H_i for $i < n$ and a surjection on H_n . For the mapping cylinder M_f we then have $H_i(M_f, Z^n) = 0$ for $i \leq n$ and $H_{n+1}(M_f, Z^n) \approx \pi_{n+1}(M_f, Z^n)$ by the Hurewicz theorem. To construct Z_{n+1} we use the following diagram:

$$\begin{array}{ccccccc}
 H_{n+1}(X) & & \pi_{n+1}(M_f, Z^n) & & H_n(X) & & \\
 \wr & & \wr & & \wr & & \\
 H_{n+1}(M_f) & \longrightarrow & H_{n+1}(M_f, Z^n) & \longrightarrow & H_n(Z^n) & \longrightarrow & H_n(M_f) \longrightarrow 0 \\
 \uparrow & & \uparrow & & \parallel & & \uparrow \\
 H_{n+1}(Z^{n+1}) & \longrightarrow & H_{n+1}(Z^{n+1}, Z^n) & \longrightarrow & H_n(Z^n) & \longrightarrow & H_n(Z^{n+1}) \longrightarrow 0
 \end{array}$$

By induction we know the map $H_n(Z^n) \rightarrow H_n(M_f) \approx H_n(X)$ exactly, namely, Z^n has generator n -cells, which are cellular cycles mapping to the given generators of $H_n(X)$, along with relator n -cells that do not contribute to $H_n(Z^n)$. Thus $H_n(Z^n)$ is free with basis the generator n -cells, and the kernel of $H_n(Z^n) \rightarrow H_n(X)$ is free with basis given by certain multiples of some of the generator n -cells. Choose 'relator' elements ρ_i in $H_{n+1}(M_f, Z^n)$ mapping to this basis for the kernel, and let the 'generator' elements $\gamma_i \in H_{n+1}(M_f, Z^n)$ be the images of the chosen generators of $H_{n+1}(M_f) \approx H_{n+1}(X)$.

Via the Hurewicz isomorphism $H_{n+1}(M_f, Z^n) \approx \pi_{n+1}(M_f, Z^n)$, the homology classes ρ_i and γ_i are represented by maps $r_i, g_i: (D^{n+1}, S^n) \rightarrow (M_f, Z^n)$. We form